

MATTHEW J. KUKLA

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EDUCATION

- The George Washington University** in progress
Mathematics, PhD *Washington, DC, USA*
- The George Washington University** expected Fall 2026
Mathematics, MA *Washington, DC, USA*
- University of Maryland** awarded May 2022
Mathematics, BSc *College Park, Maryland, USA*

PROFESSIONAL EXPERIENCE

- The Math Citadel** March 2019 - present
Academic Researcher
- Conduct original research in fuzzy algebra, stochastic geometry, graphical probabilistic models
 - Develop software packages:
 - Design and implement fuzzy anomaly detection models
 - Optimize numerical methods for modeling random processes
- BlueHalo Labs** June 2022 - May 2025
Research Engineer *Rockville, Maryland, USA*
- Researcher in mathematics with a focus on graph theory, formal logic, scientific computing
 - Designed, implemented, and deployed novel graph clustering algorithms. Optimized with high-performance linear algebra libraries.
 - Constructed systems for knowledge representation, formal ontology, and automated reasoning across large relational structures
 - Developed specialized tools for signal processing and harmonic analysis
 - Wrote research articles and technical reports for delivery to government, academic, and private-sector customers

SKILLS

Programming Languages	C, OCaml, Python, Fortran, Julia, Prolog, Java, MATLAB
Operating Systems	Linux, UNIX (BSD and Solaris), MS-DOS
Tools	Shell scripting, sed, AWK, Git, L ^A T _E X
Libraries	NumPy, SciPy, BLAS, LAPACK
Web	HTML, CSS, OWL, RDF, Gopher, AWS
Databases	SQL, Solr, ElasticSearch, Cypher
Radio	NEC2, GNURadio, SDR

PUBLICATIONS AND PREPRINTS

- Double Orthogonal Factorization Systems**
(with C.B. Aberlé, Elena Caviglia, Rubén Maldonado, Luca Mesiti, Dorette Pronk, and Tanjona Ralaivaosoana)
Preprint, arXiv:2509.26343, submitted. (2025)
- Logical Limit Laws for Layered Permutations and Related Structures**
(with Samuel Braufeld)
Published, Enumerative Combinatorics and Applications. 2 no. 4. (2021)

Colored Convex Linear Orders and Logical Limit Laws

Preprint. (2021)

Rings of Typed Ordered Fuzzy Numbers

(with Cassandra Traylor)

Preprint, arXiv:2010.07764. (2020)

CONFERENCE TALKS

Convergence Laws for Linearly-Ordered Structures and Layered Permutations

AMS Eastern Sectional Meeting, Special Session on Mathematical Logic, Washington, DC (2026)

Orthogonal Factorization Systems for Double Categories

The International Category Theory Conference, Johns Hopkins University (2026)

Relational Structures, Logical Limit Laws, and Layered Permutations

Knots in Washington 51, The George Washington University (2025)

Double Factorization Systems and Double Fibrations

7th International Conference on Applied Category Theory, University of Oxford (2024)

INVITED SEMINAR TALKS

Double Categories for the Working Graduate Student

Graduate Student Seminar, The George Washington University (2026)

First-Order Logical Limit Laws, Ordered Structures, and Permutation Classes

Computability & Complexity Seminar, The George Washington University (2025)

Logical Limit Laws for Layered Permutations and Related Structures

Mathematical Logic Seminar, University of Maryland (2022)

Categorical Mirror Symmetry of Elliptic Curves (two lecture series)

Geometry and Physics Seminar, University of Maryland (2018)

Generalized Calabi-Yau Manifolds

Geometry and Physics Seminar, University of Maryland (2018)

TEACHING

The George Washington University, Department of Mathematics

Graduate Instructional Assistant

August 2025 - present

Washington, DC, USA

GW Experimental Mathematics Lab

Lab Project Assistant

Spring 2026

Washington, DC, USA

- Mentored undergraduate research project in dynamical systems, studying confinement properties of repulsive-attractive kernels

SERVICE

Conference Organization

AMS Special Session on Computable Structure Theory

(co-organized with Valentina Harizanov, Jeremias Valenzuela Morales, and Paula de Lima Souza)

AMS Eastern Sectional, 3 October 2026

LICENSES AND CERTIFICATIONS

EPA Part 608 Universal Certification

For service of stationary HVACR equipment and handling of refrigerants.